

kinematics of rigid body

Exercise 01.

1. Represent the points A and B with polar coordinates:

$$(1,2) \text{ and } \left(2, \frac{5\pi}{4}\right);$$

2. Represent the points C and D with cylindrical coordinates:

$$\left(2, \frac{\pi}{2}, 2\right) \text{ and } (1, \pi, -1).$$

3. Express the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ in the corresponding local coordinate systems.

4. Represent the points E and F with spherical coordinates:

$$\left(2, \frac{\pi}{4}, \frac{\pi}{4}\right) \text{ and } \left(1, \frac{\pi}{3}, \frac{3\pi}{4}\right); \text{ Express the vector } \overrightarrow{EF} \text{ in the local coordinate system } (E, \vec{e}_r, \vec{e}_\theta, \vec{e}_\varphi).$$

Exercise 02.

A material point moves along a trajectory described by the

$$\text{following parametric equations: } \begin{cases} x = t \\ y = 2t^2 \\ z = 0 \end{cases}$$

Determine:

1. The unit tangent vector $\vec{u}$ to the trajectory;
2. The radius of curvature ρ
3. The normal $\vec{n}$ to the trajectory;
4. The binormal $\vec{b}$.

Exercise 03:

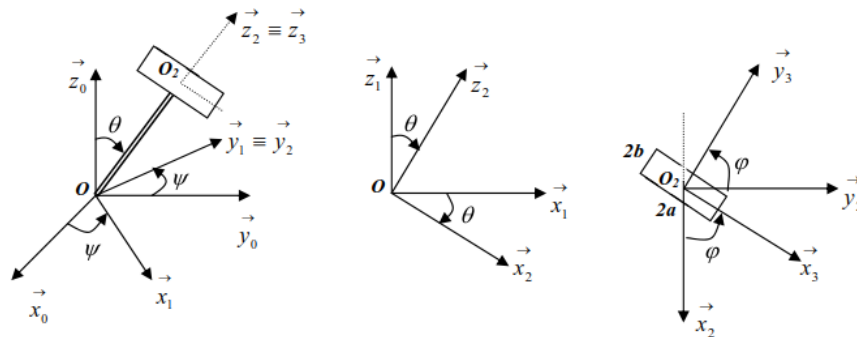
We consider a mechanical system composed of: A rigid rod (OO_2) of length L , and a rectangular plate of dimensions $2a \times 2b$, hinged at O_2 to the rod (see figure).

R_0 is the fixed reference frame; R_1 rotates by an angle ψ about the axis $\vec{z}_0$. The plate rotates about the rod with angular velocity $\dot{\varphi}$.

Given: $\dot{\psi} = \text{Const}$; $\dot{\theta} = \text{Const}$ and $\dot{\varphi} = \text{Const}$.

Determine:

1. The transformation matrices from R_1 to R_2 ;
2. The instantaneous rotation vector of R_3 relative to R_0 expressed in R_2 ;
3. The velocity $\vec{V}^0(O_2)$ expressed in frame R_2 by differentiation;
4. The velocity $\vec{V}^0(A)$ with respect to R_0 expressed in R_2 by the solid's kinematics;
5. The acceleration expressed $\vec{\gamma}^0(O_2)$ in frame R_2 by differentiation and using rigid body kinematics;



Exercise 04:

Two bars are hinged at point A and are part of a control mechanism.

Bar OA rotates about the axis $\vec{z}_0$ in the horizontal plane (x_0, y_0) .

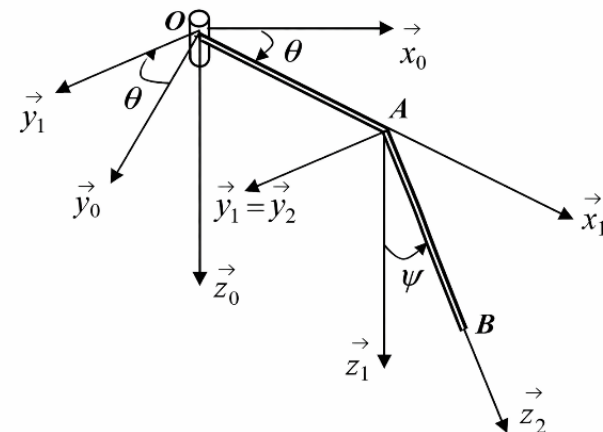
Bar AB rotates about the axis $\vec{y}_1$ in the plane (x_1, z_0)

Let P be a moving point on bar AB such that: $\overrightarrow{AP} = r\vec{z}_2$, $\overrightarrow{OA} = a\vec{x}_1$, $\overrightarrow{AB} = b\vec{z}_2$; where a and b are constants.

R_1 is the projection frame

Determine:

1. The transformation matrices from R_0 to R_1 , and from R_2 to R_1 ;
2. The instantaneous rotation vector $\vec{\Omega}_2^0$, the velocity $\vec{V}^0(B)$, and the acceleration $\vec{\gamma}^0(B)$, by direct differentiation and using rigid body kinematics.



Exercise 05.

A fan is modeled by the system shown in the figure opposite . It consists of an arm OAB , bent at A , and a propeller with center B . The arm undergoes a rotational motion about the axis $\vec{z}_0$. The propeller rotates about the axis AB .

Given:

$$R_2 // R_1; OA = a; AB = b, BC = r$$

$$\dot{\phi}(t) = \text{Const}; \dot{\beta} = \text{Const}$$

Taking $R_2(B, \vec{x}_2, \vec{y}_2, \vec{z}_2)$ as the projection reference frame, Determine:

1. The rotation vector of the propeller with respect to frame R_0 .
2. The velocity and acceleration vectors of point B with respect to R_0 , using the differentiation method.
3. The velocity and acceleration vectors of point C with respect to R_0 , using rigid body kinematics.

